

Maths Behind Ridge Regression

$$L = (y_i - mx_i - b)^2 + \lambda m^2$$

$$b = \bar{y} - m\bar{x}$$

Hence,

$$L = \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x})^2 + \lambda m^2$$

$$\frac{\partial L}{\partial m} = 2 \sum (y_i - mx_i - \bar{y} + m\bar{x}) (-x_i + \bar{x}) + 2\lambda m = 0$$

$$= -2 \sum_{i=1}^n (y_i - \bar{y} - mx_i + m\bar{x})(x_i - \bar{x}) + 2\lambda m = 0$$

$$= \lambda m - \sum_{i=1}^n [(y_i - \bar{y}) - m(x_i - \bar{x})](x_i - \bar{x}) = 0$$

n

$$= \lambda m - \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2 = 0$$

$$= \lambda m - \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + m \sum_{i=1}^n (x_i - \bar{x})^2 = 0$$

$$= \lambda m + m \sum_{i=1}^n (x_i - \bar{x})^2$$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2 + \lambda} \rightarrow \text{Ridge Regression}$$

$\lambda = \text{Hyperparameter}$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \text{ Linear Regression}$$

Ridge Regression for nD Data.

$w_0 \ w_1 \ w_2 \ \dots \ w_n$.

$$\begin{array}{c|c} x_1 & x_2 & \dots & x_n & y \end{array} \Rightarrow n+1 \text{ columns}$$

$$L = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

Converting to matrix

$$= (XW - Y)^T (XW - Y)$$

$\therefore$

$$Y = \begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix} \quad W = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{m1} & x_{m2} & & x_{mn} \end{bmatrix}$$

Loss function for LR

$$L = (XW - Y)^T (XW - Y)$$

Loss function for RR

$$L = (XW - Y)^T (XW - Y) + \lambda \|W\|^2$$

In matrix form

$$L = (xW - y)^T (xW - y) + \lambda W^T W$$

$$= [(xW)^T - (y)^T] (xW - y) + \lambda W^T W$$

$$= (W^T x^T - y^T) (xW - y) + \lambda W^T W$$

$$= W^T x^T W - W^T x^T y - y^T x W + y^T y + \lambda W^T W$$

$$= W^T x^T W - 2W^T x^T y + y^T y + \lambda W^T W$$

$$\frac{dL}{dW} = 2x^T x W - 2x^T y + 0 + 2\lambda W = 0$$

$$x^T x W + \lambda W = x^T y$$

$$(x^T x + \lambda I) W = x^T y$$

$$\therefore W = (X^T X + \lambda I)^{-1} X^T Y$$

$$W = (X^T X)^{-1} X^T Y$$